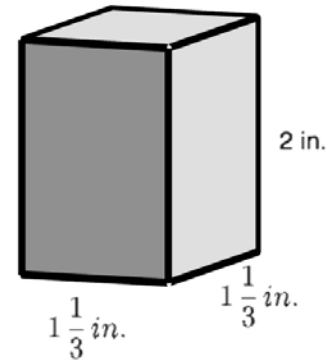


Grade 6 Accelerated
Unit 8
Lesson 2 Homework

Name _____
Date _____
Period _____

A rectangular prism is shown with units measured in inches.

In questions 1–4, find the volume using cubes with a side length of $\frac{2}{3}$ in.



1. Determine how many cubes will fit along the length, width, and height.

$$1\frac{1}{3} \div \frac{2}{3} = \underline{\hspace{2cm}} \text{ cubes will fit along the length.}$$

$$1\frac{1}{3} \div \frac{2}{3} = \underline{\hspace{2cm}} \text{ cubes will fit along the width.}$$

$$2 \div \frac{2}{3} = \underline{\hspace{2cm}} \text{ cubes will fit along the height.}$$

2. A total of _____ cubes will fit within the rectangular prism.
3. The volume of each cube is $\frac{8}{27} \text{ in}^3$. Multiply the number of cubes needed to fill the prism by the volume of each cube in order to find the volume of the prism. Include units.
4. Instead of using cubes, Kaia uses the formula $V = l \times w \times h$. Fill in the spaces below to complete her work. Include units.

$$V = \underline{\hspace{2cm}} \times \underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$$

$$V = \underline{\hspace{2cm}} \text{ in.}^3$$

A baker is packaging her cupcakes into single serve boxes for a special event. Each cube-shaped box has side lengths 3.5 inches (in.). The smaller boxes are then packed into a larger box with dimensions of 10.5 in. $\times$ 7 in. $\times$ 7 in.

5. How many single serve cupcakes can be packed into the larger box?
6. Use the number of cupcakes that can be packed in the larger box to determine the volume of the box. Include units.

Owen and Noah are calculating the volume of a rectangular prism with the dimensions of 2 cm $\times$ 2 cm $\times$ 1 cm using different size cubes. Noah uses cubes with side lengths $\frac{1}{3}$ cm and Owen uses cubes with side lengths $\frac{1}{6}$ cm.

7. Who will need more cubes to fill the larger prism? Justify.
8. What is the volume of the rectangular prism? Include units.
9. If they had used a cube with side lengths $\frac{1}{4}$ cm instead, would they have found the same volume of the rectangular prism?
10. An aquarium in the shape of a rectangular prism is being filled with water. The dimensions of the aquarium are 48.5 inches (in.) by 18.25 in. If the height of the water is 15.75 in., what is the volume of the water in cubic inches?